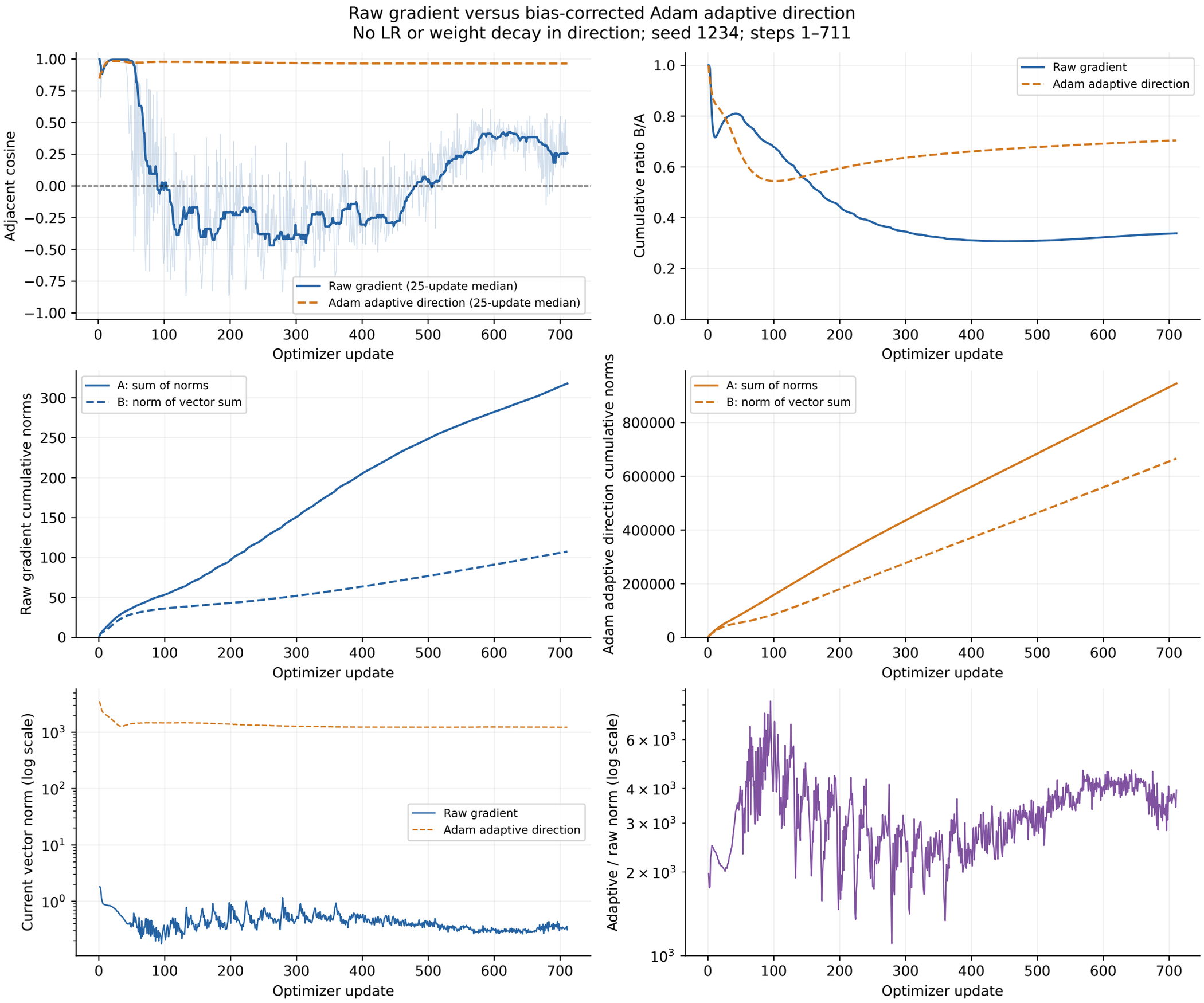

Raw gradient versus bias-corrected Adam adaptive direction
No LR or weight decay in direction; seed 1234; steps 1–711
Adjacent cosine
Raw gradient (25-update median)
Adam adaptive direction (25-update median)
Optimizer update
Cumulative ratio B/A
Raw gradient
Adam adaptive direction
Optimizer update
Raw gradient cumulative norms
A: sum of norms
B: norm of vector sum
Optimizer update
Adam adaptive direction cumulative norms
A: sum of norms
B: norm of vector sum
Optimizer update
Current vector norm (log scale)
Raw gradient
Adam adaptive direction
Optimizer update
Adaptive / raw norm (log scale)
Optimizer update